Capstone Design Project I

Progress Report



A Modular Self-Reconfigurable Robot (MSRR) for Search and Space Exploration

Submitted to the
Project Committee appointed by the
International School of Engineering (ISE)
Faculty of Engineering, Chulalongkorn University

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1/2026: 2147498 Capstone Design Project I
Robotics and Artificial Intelligence Engineering (International Program)
International School of Engineering (ISE), Faculty of Engineering,
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1 Project Timeline

- **Completed:** Project concept, objectives, literature review, and phased system architecture established.
- **In progress:** Phase A — single-module locomotion, sensing, communication, and simulation baseline.
- **Next:** Phase B — EP-Face docking, task-oriented reconfiguration, and simple assembled locomotion.

2 Project Phases

2.1 Phase A

External perception and computation. Develop the mechanically homogeneous module, differential-drive locomotion, joint control, odometry, and module-to-commander communication. Validate the module and physics twin before physical docking is introduced.

2.2 Phase B

Docking and task-oriented reconfiguration. Add the electropermanent EP-Face connectors, infrared alignment, and connector-state feedback. Extend the reconfiguration pipeline with task assignment, parallel assembly sequencing, and selection from a library of feasible configurations.

2.3 Phase C

Onboard perception with external computation. Mount a monocular RGB camera and an 8×8 multi-zone ToF sensor on the leader module. Estimate obstacle geometry and traversability locally, while transmitting sensor data to an external computer for processing.

2.4 Phase D

Fully onboard perception and computation. Move the perception and configuration-selection pipeline onto a coordinator module using an embedded compute board. Close the perceive–select–reconfigure–traverse loop on hardware without external localisation or computation.

3 Gantt Chart

Task schedule from the team Notion timeline (as of 25 Sep 2026), grouped by phase.

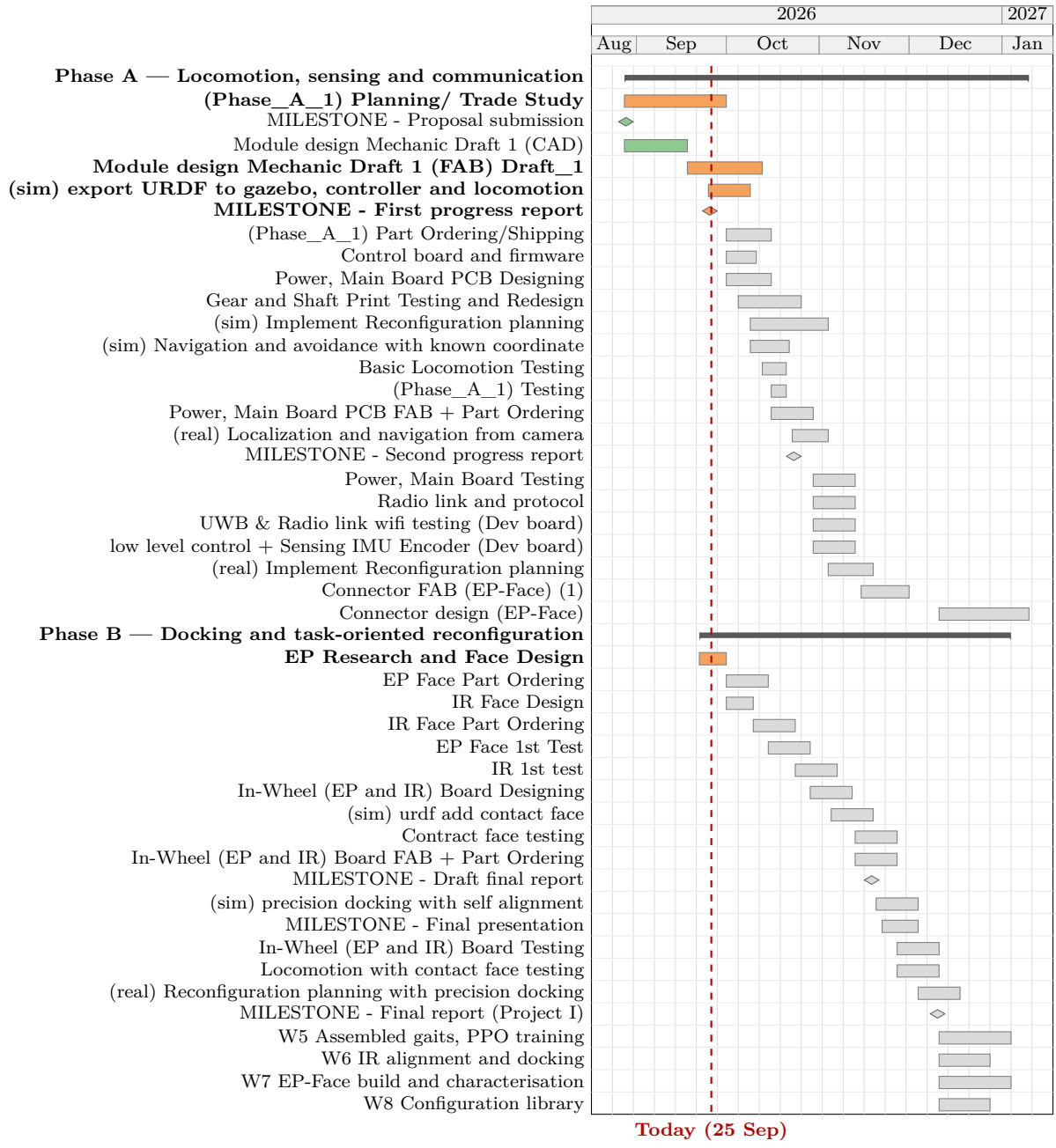


Figure 1: Project I schedule (Phases A–B). Completed In progress Upcoming; dashed line marks the report date.

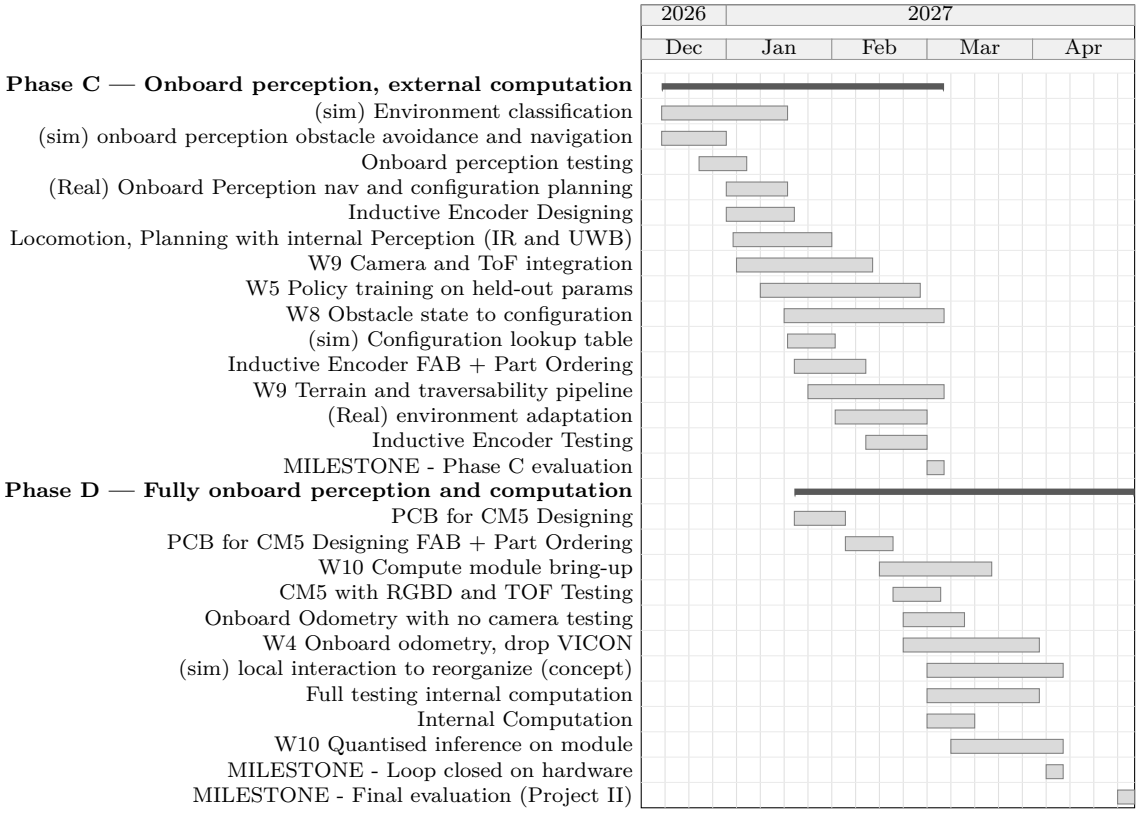


Figure 2: Project II schedule (Phases C–D). All tasks upcoming (▭).

4 Module CAD Design

The design is based on SMORES-EP [1] (originally 80 mm), scaled to a $100 \times 100 \times 100$ mm envelope for additional packaging space.

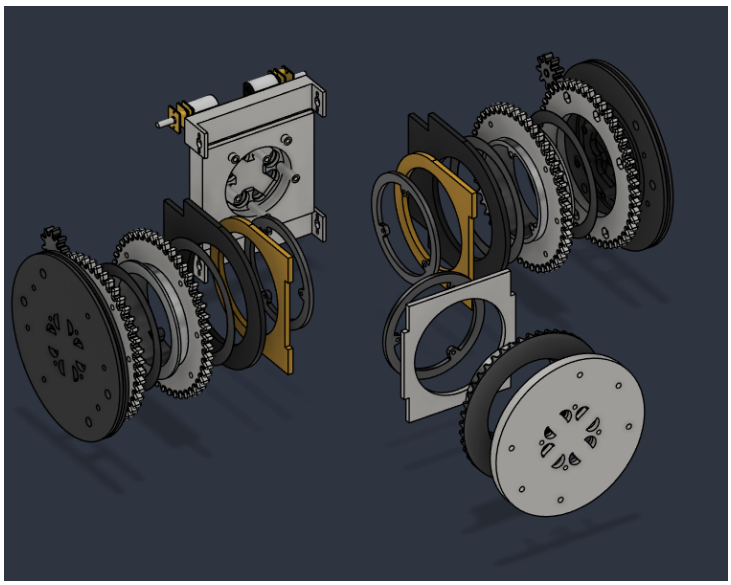
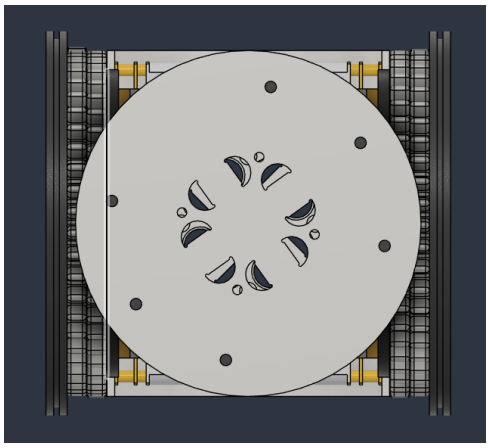
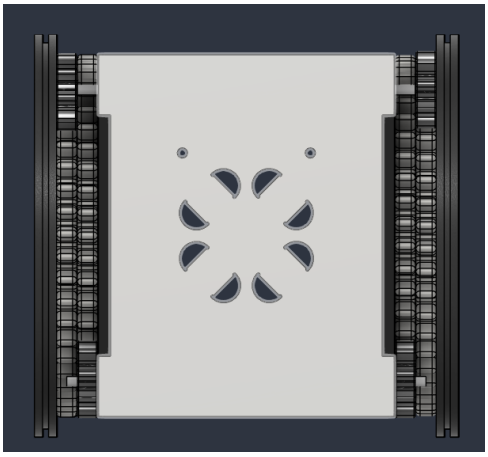


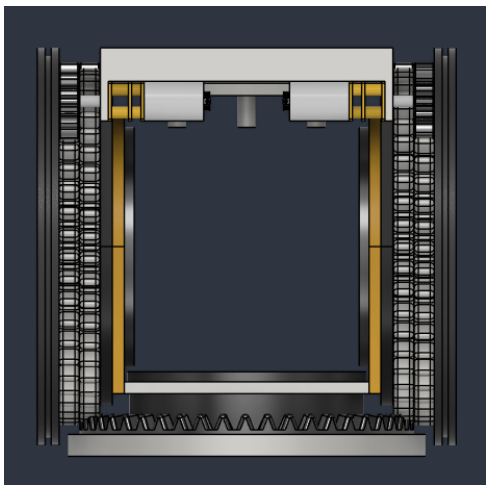
Figure 3: Exploded view of the module.



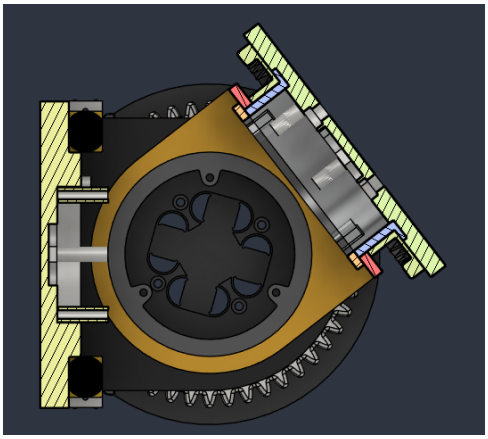
(a) Front view.



(b) Rear view.



(c) Top view.



(d) Tilted sectional view.

Figure 4: Module CAD views.

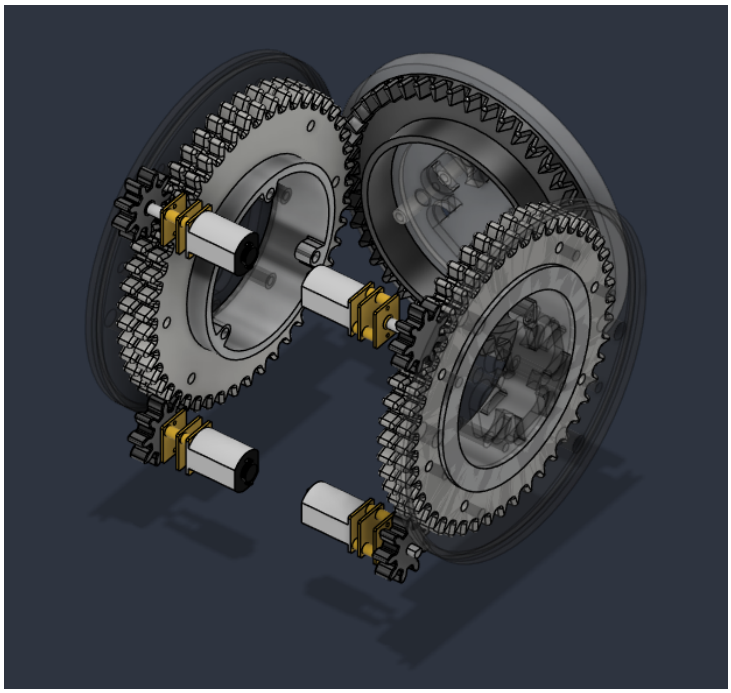


Figure 5: CAD view of the gear train and motor arrangement.

4.1 Gear Train Design

4.1.1 SMORES-EP Reference Baseline

The original SMORES-EP gear train was reverse-engineered from published data (Table 1), yielding module $m = 1.4$, pitch diameter 67.2 mm, and stage ratio $48/9 = 5.33$.

Table 1: Reconstructed SMORES-EP reference parameters.

Parameter	**SMORES-EP**
Module envelope	$80 \times 80 \times 80$ mm
Spur gear	48 teeth
Crown gear	Same part as spur, used at 90°
Pinion	9 teeth
Gearbox	298:1

4.1.2 New Gear Geometry

The 100 mm envelope allows a max spur gear OD of 90 mm. The 5.33:1 ratio was preserved (64/12 teeth) to keep the gearbox selection, SMORES-EP benchmarks, and control firmware valid. The 64/12 pair was chosen over 48/9 because 12 teeth reduces undercutting, improves contact ratio, and provides a larger pinion bore.

Unlike SMORES-EP (which reuses a spur gear as a crown gear), this design uses a **true crown (face) gear** for better load distribution. The crown gear was sized to 84 mm OD (46 teeth, $m = 1.75$) after a 90 mm first attempt interfered with surrounding components.

This creates a **module mismatch** ($m = 1.36$ spur vs. $m = 1.75$ crown) — an accepted compromise since both diameters and tooth counts are fixed by packaging and ratio constraints. The crown dedendum was reduced to 2.15 mm (from standard 2.19 mm) to clear tip interference.

Table 2: Final gear train parameters.

Component	**Teeth**	**Module**	**OD**	**Notes**
Wheel spur gear	64	1.36	90 mm	5 mm face width, bolt holes in web
Pinion (both)	12	1.36	19.04 mm	Through-bore (pan/tilt) or blind (wheel)
Crown gear	46	1.75	84 mm	Dedendum 2.15 mm; true face-gear

Common: 20° pressure angle, 0.2 mm backlash, 0.3 mm root fillet, 0.1 mm bore tolerance. All gears generated via Fusion 360. Material: 3D-printed nylon.

4.1.3 Drivetrain Architecture

Four motors drive four spur gears:

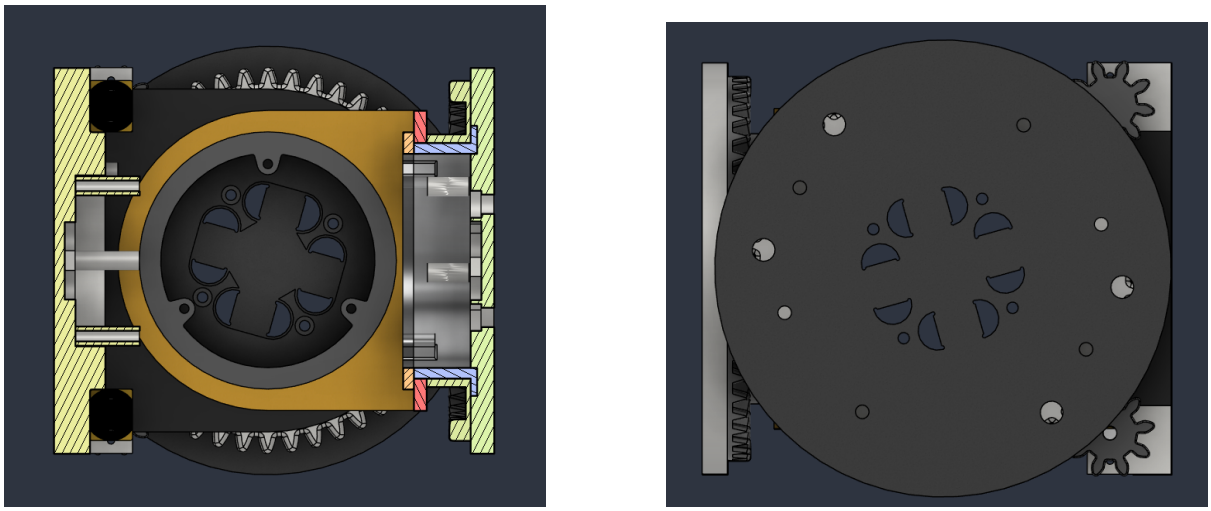
- **Outer two** — each drives a wheel (differential-drive locomotion).
- **Inner two** — both mesh with a single crown gear: same direction → **pan**; opposite directions → **tilt**.

4.1.4 Status

All geometry is CAD-only. **No parts have been printed or tested.** Priority next steps: print first articles, verify spur/crown mesh despite module mismatch, check 12-tooth pinion undercut at print tolerances, and validate backlash and bore fit.

4.2 Outer Wheels

Two independently driven wheels (100 mm diameter) with NBR N70 O-ring tire groove (1.5 mm cross-section), EP-magnet docking space, and PCB mounting posts.



(a) Interior view (wheel and drivetrain). **(b)** Exterior view (wheel and module).

Figure 6: Outer wheel CAD.

4.3 Pan Face

90 mm diameter, no O-ring groove. Includes EP-magnet docking space.

4.4 Tilt Face

Approximately 90 × 71.6 mm rectangular part with EP-magnet docking space and an additional motor mount.

5 Electronics

5.1 Phased Electronics Plan

Electronics are built incrementally across project phases:

- **Phase A-1:** COTS parts — 2S battery, 4-channel motor driver with MCU, ESP32-C3 for Wi-Fi.
- **Phase A:** Custom PCBs — power board (buck, LDO, 4× H-bridge with current sensing) and main board (STM32G474, IMU, UWB, Wi-Fi).
- **Phase B:** In-wheel PCB with EP-Face circuitry and IR contact-alignment, connected via slip ring.
- **Phase C:** RGB-D/ToF camera (Sipeed MaixSense A075V) + wheel encoder added; compute still external.
- **Phase D:** Raspberry Pi CM5 on custom carrier PCB moves compute onboard.

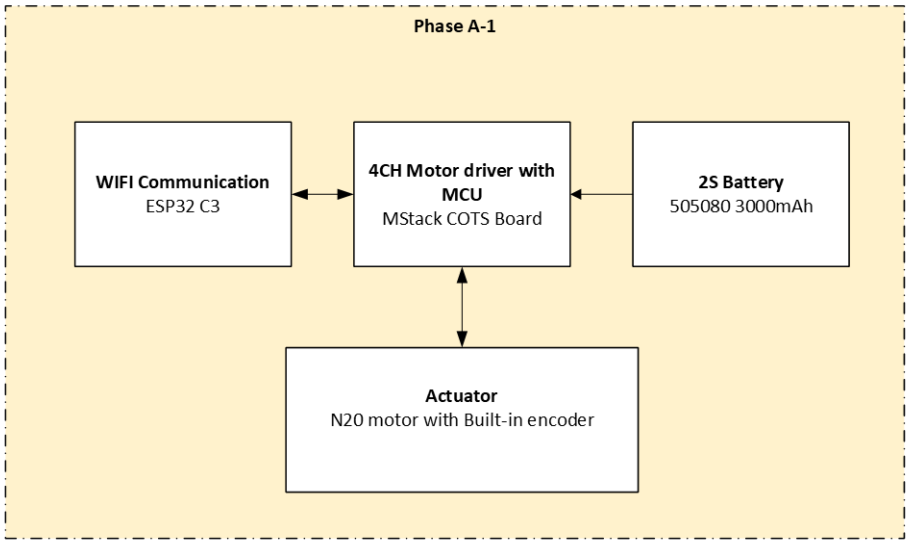
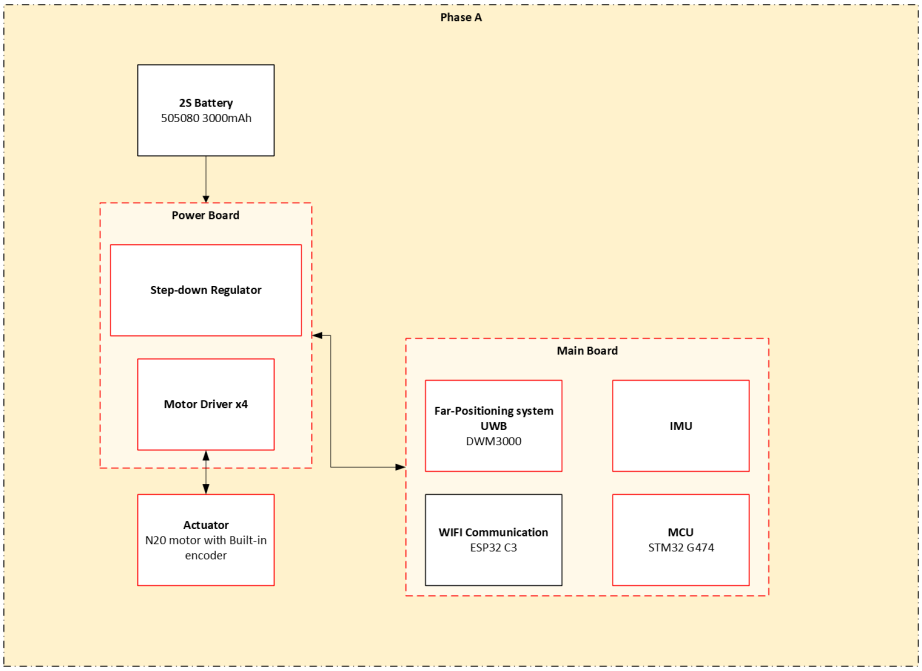
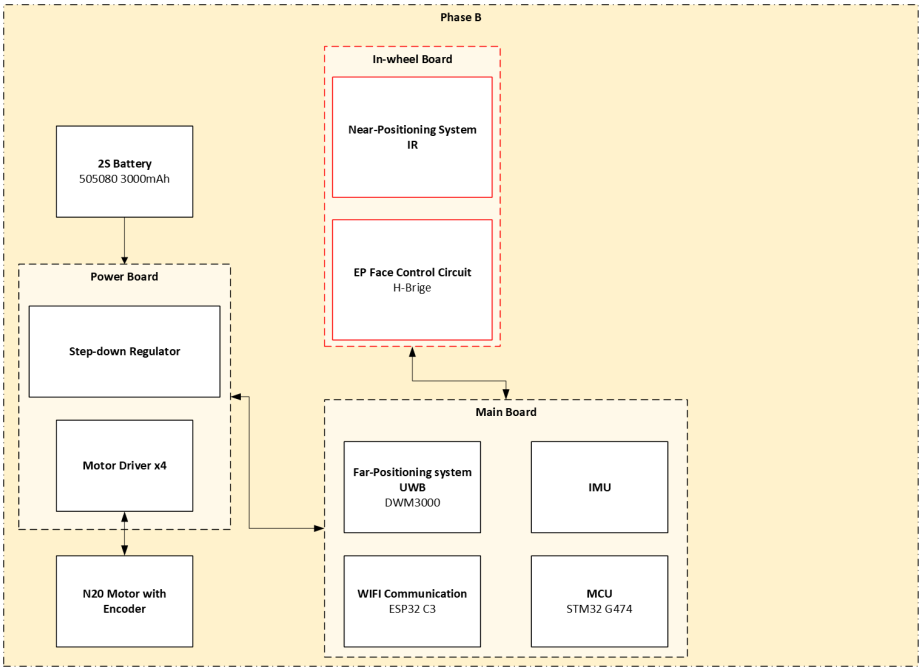


Figure 7: Phase A-1 electronics block diagram.

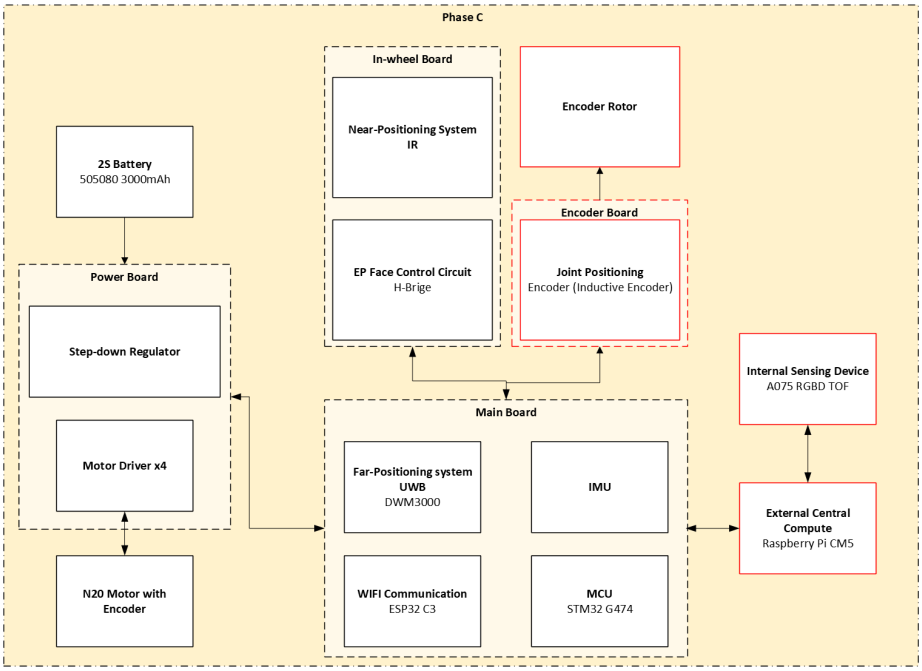


(a) Phase A.

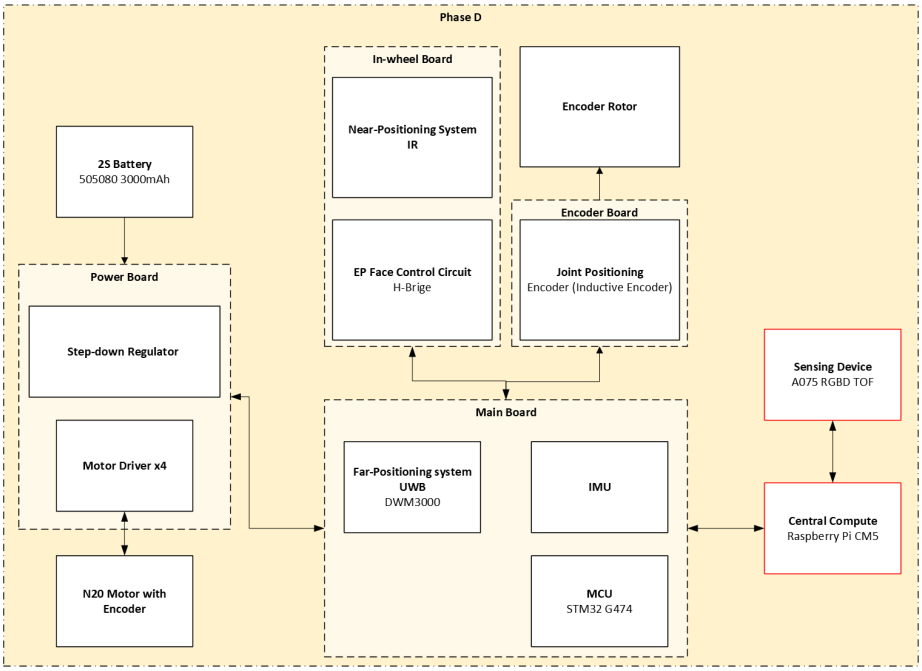


(b) Phase B.

Figure 8: Electronics block diagrams, Phases A–B.



(a) Phase C.



(b) Phase D.

Figure 9: Electronics block diagrams, Phases C–D.

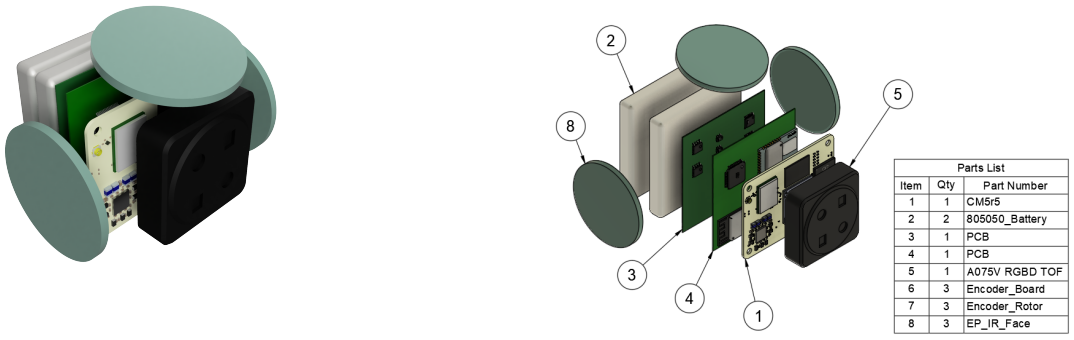
5.2 Component Selection

Table 3: Key electronics components.

Component	Phase	Key Specs	Interface / Supply
4× N20 gearmotor	A-1+	6 V, 50 rpm; 7 PPR encoder	A/B quadrature; H-bridge
STM32G474	A+	Cortex-M4F @ 170 MHz; 12-bit ADC; FDCAN	SPI/I2C/UART/CAN; 3.3 V
ICM-42688-P	A+	±2000 dps / ±16 g; 32 kHz ODR	SPI or I2C; 1.8 V
DW3000 (UWB)	A+	IEEE 802.15.4z; ∼10 cm ranging	SPI; 3.3 V
ESP32-C3	A-1+	Wi-Fi 4 + BLE 5; RISC-V @ 160 MHz	UART/SPI; 3.3 V
EP-Face + IR PCB	B	Custom; EP magnet + IR alignment	Slip ring; 3.3 V
RPi CM5	D	Cortex-A76 @ 2.4 GHz; up to 16 GB RAM	Custom carrier; 5 V
Sipeed MaixSense A075V	C+	ToF 320×240 @ 30 fps; RGB 800×600	USB 2.0; 5 V

5.3 Electronics Packaging

All components fit within the 100 mm cube based on rough CAD placement (Figure 10).



(a) Assembled. **(b)** Exploded.

Figure 10: Electronics packaging.

5.4 Wheel Encoder

Replacing the original continuous-rotation potentiometer (∼5° resolution, contact wear) with a non-contact encoder. Three options considered:

Table 4: Wheel encoder options.

	On-axis magnetic (AS5600)	**Off-axis tape**	**Custom inductive**
Output	Absolute, 12–14 bit	Incremental + index	Absolute
EP interference	Sensitive	Sensitive	Immune
Design effort	Low	Medium	High

The inductive option best handles the EP connector's nearby field but has the highest schedule risk. Decision deferred to Phase C.

5.5 Changes from Progress Report 1

The phased plan (A-1 to D) and its scope are unchanged. The electronics work plan is delayed by one week, which shifts the electronics timeline and the work that depends on it by the same amount.

Phase A-1 was built from an STM32F411 development board, two TB6612FNG dual H-bridge drivers and an ESP32-C6 module, instead of the COTS 4-channel motor driver with a built-in MCU and the ESP32-C3 described in PR1. This gives direct control of the PWM, direction and encoder signals, so the wiring and firmware carry over to the Phase A main board.

The only other hardware change is the battery. The 2S 3000 mAh 505080 pack was replaced with a 3S pack of Vapcell M11 18350 Li-ion cells (1100 mAh, 10 A continuous, 15 A burst), which can supply more current for the EP-Face in Phase B. The smaller capacity lowers the estimated run time (Section 5.7).

5.6 Phase A-1 Hardware

Phase A-1 is the bench version of the locomotion electronics. It tests the wiring, power path and firmware set-up that will move onto the Phase A PCBs.

5.6.1 Hardware Schematic

Figure 11 shows the Phase A-1 wiring. An STM32F411CEU6 development board controls four N20 gearmotors through two dual-channel TB6612FNG H-bridge drivers (two motors each, shared standby line). Each motor has a PWM input and two direction inputs, and its quadrature encoder is read back by the MCU. M1 and M2 drive the left and right wheels; M3 and M4, the inner-left (IL) and inner-right (IR) motors, drive pan and tilt. An ESP32-C6 module provides the Wi-Fi link over UART, an ST-Link handles SWD programming, and USB gives a serial console for testing.

Three buck converters run from the 3S pack through an XT30 connector. A 5 V buck supplies the MCU board and the ESP32-C6; the MCU board's 3.3 V regulator supplies the driver logic and the encoders. One 6 V buck per TB6612FNG supplies the motors, so the N20s run at their rated voltage at full PWM duty. A 100 nF capacitor across each motor suppresses brush noise, and a 10 kΩ pull-down on STBY keeps both drivers disabled until the MCU enables them.

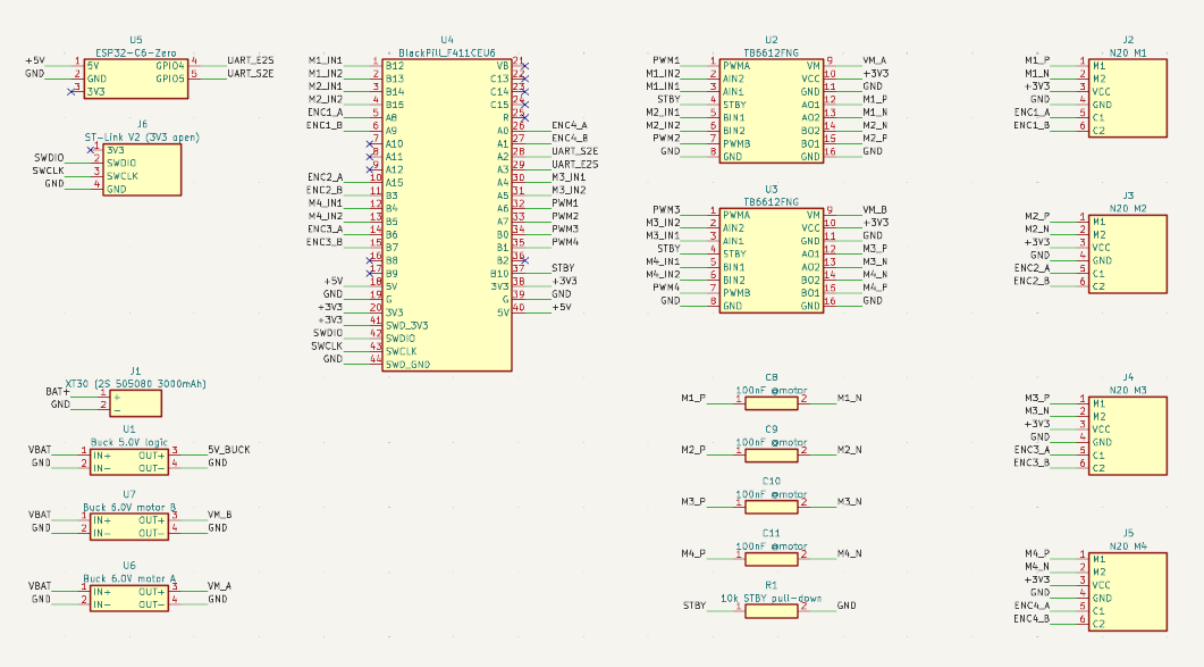
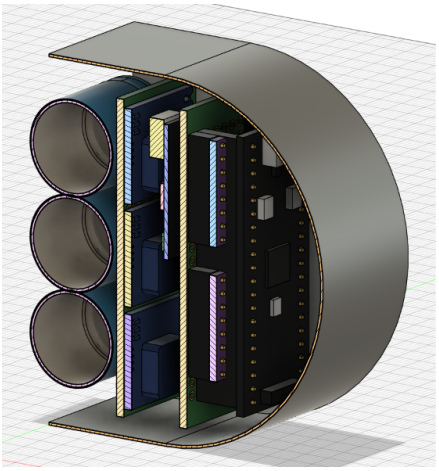


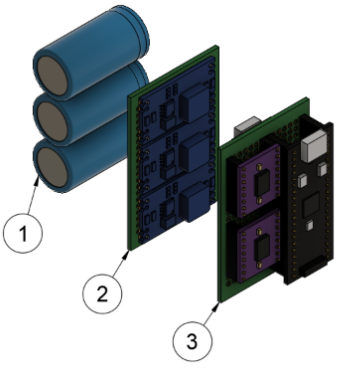
Figure 11: Phase A-1 hardware schematic.

5.6.2 Hardware Packaging

Figure 12 shows the Phase A-1 hardware placed in the module. Three parts stack front to back inside the module shell. The three 18350 cells sit in a vertical column at the back. In front of them is the regulator board with the three buck converters. The controller board, carrying the STM32F411 board, both TB6612FNG drivers and the ESP32-C6, sits at the front.



(a) Assembled.



Parts List			
Item	Qty	Part Number	Material
1	1	Battery_Pack	Steel
2	1	MSRR_A1_board2_regulators	Steel
3	1	MSRR_A1_board1_esp_flat	Steel

(b) Exploded.

Figure 12: Phase A-1 hardware packaging.

5.6.3 Power Calculation

Table 5 gives the Phase A-1 load on the 3S pack, using the PR1 load figures. Motor power passes through the 6 V regulators (90 %) and the TB6612FNG (95 %); logic power through a 5 V buck (90 %) and the board's 3.3 V LDO (66 %).

Table 5: Phase A-1 power budget (3S pack, 9.8 Wh usable).

Load	Typical	All motors driving	Peak (all stalled)
4× N20 via 6 V reg. + TB6612	2.6 W[a]	7.0 W	28.1 W
STM32F411 + ESP32-C6 (3.3 V)	0.7 W	0.7 W	2.2 W[b]
Total from pack	**3.3 W**	**7.7 W**	**30.3 W**
Pack current (11.1 V)	0.30 A	0.69 A	2.7 A (3.4 A at 9 V)
Run time	**3.0 h**	**1.3 h**	–

[a]PR1 duty cycle: wheels 60 %, pan/tilt 15 %, 0.25 A each loaded. [b]Wi-Fi transmit burst, 335 mA (PR1 ESP32-C3 figure).

The 3.4 A worst-case pack current is well inside the cell's 10 A continuous rating. Because the drivers see 6 V, a stalled N20 draws about 1 A, inside the TB6612FNG's 1.2 A continuous rating per channel. Each 6 V regulator must supply up to 2 A with both of its motors stalled.

5.6.4 Embedded System Setup

The STM32F411 was configured in STM32CubeMX (Figure 13, Table 6). The 25 MHz crystal is multiplied by the PLL to a 96 MHz system clock, which also gives the exact 48 MHz needed for USB. One timer generates the PWM for all four motors at 20 kHz, which is above hearing range. Each encoder has its own timer in quadrature (4×) mode with input filtering and pull-ups. A separate timer interrupt at 1 kHz sets the control-loop rate. The ESP32-C6 link uses DMA, with a circular receive buffer so incoming commands are not lost while the CPU is busy.

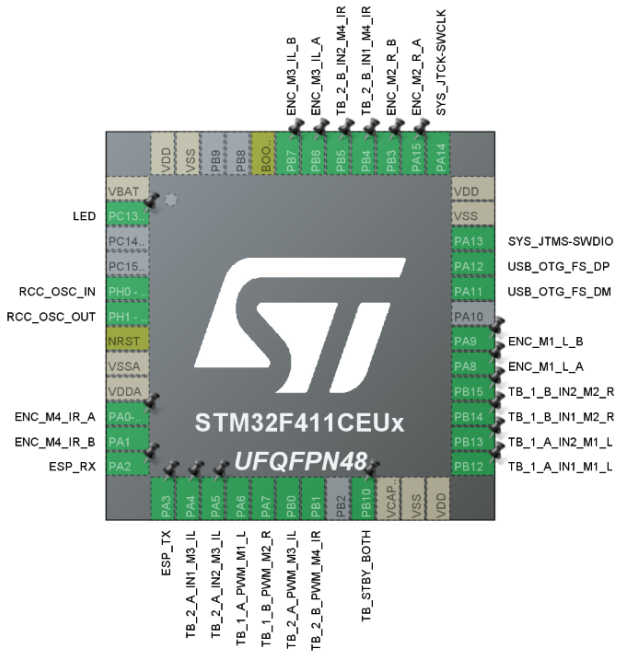


Figure 13: STM32CubeMX pin and peripheral configuration.

Table 6: STM32F411 peripheral assignment.

Function	Peripheral and pins	Setting
System clock	HSE 25 MHz → PLL	96 MHz
Motor PWM, M1–M4	TIM3 CH1–4 (PA6, PA7, PB0, PB1)	20 kHz
Motor direction	GPIO, 2 per motor (PB12–15, PA4–5, PB4–5)	–
Driver standby	GPIO PB10 (both TB6612)	–
Encoder M1 (L)	TIM1 (PA8, PA9), 16-bit	×4 quadrature
Encoder M2 (R)	TIM2 (PA15, PB3), 32-bit	×4 quadrature
Encoder M3 (IL)	TIM4 (PB6, PB7), 16-bit	×4 quadrature
Encoder M4 (IR)	TIM5 (PA0, PA1), 32-bit	×4 quadrature
Control loop tick	TIM10 interrupt	1 kHz
ESP32-C6 link	USART2 (PA2, PA3) + DMA	115 200 baud
PC console	USB OTG FS, CDC class	–
Debug	SWD (PA13, PA14)	–

5.7 New Power Calculation

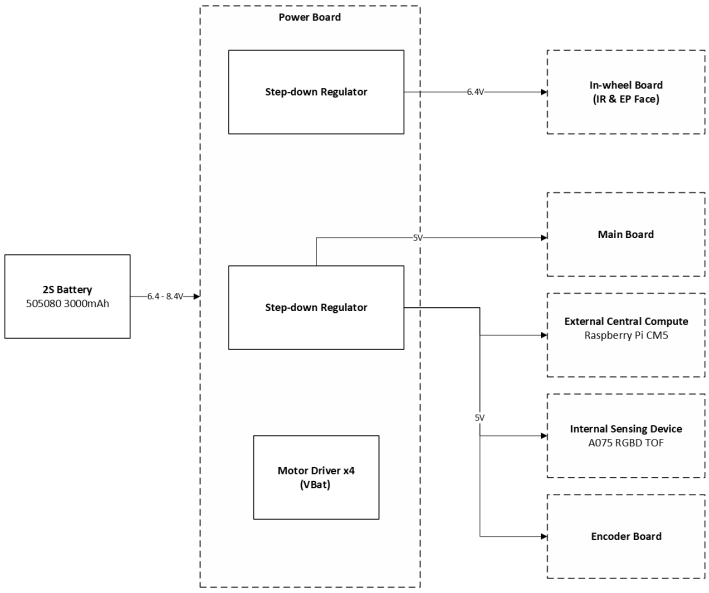


Figure 14: Power distribution block diagram.

The PR1 load budget is unchanged; only the pack changes. The Phase A power board keeps the PR1 architecture: the H-bridges run directly from the pack, not from the 6 V regulator used on the Phase A-1 bench set-up. This removes the regulator loss and a 4 A regulator from the board. Power from the 3S pack splits three ways:

- **Motors:** 4× N20 via H-bridges, PWM capped at ∼48 % (6/12.6 V at full charge) for the 6 V rating.
- **EP-Face:** Dedicated buck regulator isolates pulse transients from the logic rail.

- **Logic/compute:** Buck to 5 V (CM5, camera); LDO to 3.3 V (MCU, sensors, radios).

Table 7: Battery comparison, PR1 vs. PR2 (80 % of energy usable).

	PR1: 2S 505080	**PR2: 3S Vapcell M11**
Voltage (nominal / full / cutoff)	7.4 / 8.4 / 6.6 V	11.1 / 12.6 / 9.0 V
Capacity / energy (usable)	3000 mAh / 22.2 Wh (17.8)	1100 mAh / 12.2 Wh (9.8)
Run time, Phase A–B (∼3.1 W)	5.8 h	**3.2 h**
Run time, Phase D (∼11.5 W)	1.5 h	**0.85 h**
Peak current (∼47 W)	7.1 A (2.4 C)	5.2 A (4.7 C, cell rated 10 A)
Motor PWM cap for 6 V	∼70 %	∼48 %
H-bridge on-time stall current	1.4 A/ch	2.1 A/ch

Table 8: Maximum load per rail.

Rail	**Max Current**	**Max Power**
Motor drive (PWM to 6 V)	4.0 A (6 V eq.); 2.1 A/ch at 12.6 V	24 W
5 V logic/compute	4.0 A	20 W
3.3 V (from 5 V LDO)	0.45 A	1.5 W
EP-Face + IR	TBD	TBD
Pack total (peak)	∼4.2 A @ 11.1 V; ∼5.2 A @ 9.0 V	∼47 W

Run time roughly halves, which is acceptable for lab testing in Phases A–C. The higher pack voltage sets new ratings for the power board: H-bridges for at least 2.1 A per channel and a supply above 12.6 V with margin, and buck regulators with a 9–12.6 V input range.

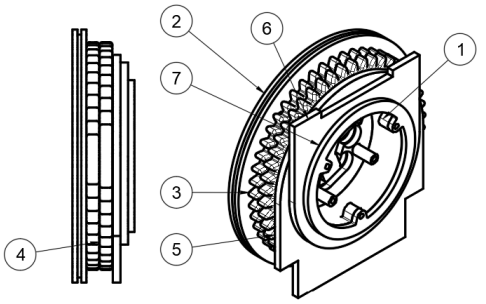
5.8 Next Steps

The tested Phase A-1 wiring and CubeMX configuration are carried over to design the Phase A main board and power board PCBs, with:

- H-bridges and buck regulators rated for the 3S pack (Section 5.7).
- The Phase A-1 pin map (Table 6) as the starting point for the main-board pin-out.

6 Shaft Tolerance Analysis (∅55 mm Bearing)

The plain bearing (OD = 55 mm) acts as the shaft. All nylon-printed components have bores around it.



Parts List		
Item	Qty	Part Name
1	1	Wheel plain bearing v6
2	1	Wheel (left and right) v19
3	2	Spur gear v17
4	2	Shim Ring 55 68 0.2 v2
5	1	Lower-tilt-joint v12
6	1	Upper-tilt-joint v8
7	1	Bushing block v3

Figure 15: Shaft and bearing component arrangement.

Table 9: Joint classification and ISO fits for ∅55 mm bearing.

Joint	Component	Motion	Fit	ISO Class
J1	Wheel + 1st Spur Gear	Dynamic	Clearance	F7/h6
J2	2nd Spur Gear (Pan/Tilt)	Dynamic	Clearance	F7/h6
J3	Upper Tilt Joint	Dynamic	Clearance	F7/h6
J4	Lower Tilt Joint	**Static**	**Interference**	P7/h6

ISO 286 limits (bearing h6: 55.000–54.981 mm): Dynamic F7 bore 55.030–55.060 mm (clearance +0.030 to +0.079 mm); Static P7 bore 54.938–54.968 mm (interference +0.013 to +0.062 mm).

FDM nylon compensation: Printed holes shrink ∼0.10–0.30 mm. Dynamic bores get +0.20 mm compensation; static bores get +0.10 mm (shrinkage helps create interference).

Table 10: Recommended CAD bore sizes.

Part	Joint	CAD Bore (mm)	Expected Fit
Wheel / Spur Gears / Upper Tilt	J1–J3 Dynamic	∅55.30	≈ 0.10–0.20 CL
Lower Tilt Joint	J4 **Static**	∅54.90	≈ 0.05–0.15 IF

CL = Clearance IF = Interference Bearing OD = ∅55.00 mm

Test plan: Print nylon test rings at five bore sizes per fit type (dynamic: 55.10–55.50 mm; static: 54.90–55.30 mm). Cool ≥2 h, measure with caliper at 3 angles. Pick smallest dynamic bore that spins freely; largest static bore that prevents rotation.

7 Software Testing Plan (Phase A)

7.1 Objectives

Verify the integrated software pipeline: AprilTag detection and robot ID via external camera → real-time pose estimation → PID motor control → path planning and execution for MSRR configurations.

7.2 Setup

MSRR robot in a controlled indoor environment with an external camera, AprilTags for ID and pose reference, and a computing system running perception, localisation, planning, and control.

7.3 Goal Configurations

Snake, Car, DoubleDriver, L-shape.

7.4 Evaluation Metrics

- **Detection rate** — % frames with correct AprilTag ID.
- **Position / orientation error** — estimated vs. reference pose.
- **Tracking error** — desired vs. actual position and heading during path following.
- **Control response** — PID overshoot and steady-state error.
- **Processing rate** — perception + control update frequency.

7.5 Acceptance Criteria

Reliable tag detection, acceptable localisation and tracking error for real-time control, stable PID behaviour without excessive oscillation, and continuous operation without critical failures. Specific numerical thresholds to be defined based on system requirements.

8 Preparation for Phase B: EP Magnet

The docking system uses an Electro-Permanent Magnet (EPM) connector based on SMORES-EP [1]. Each EP unit pairs an AlNiCo 5 magnet (low coercivity, switchable) with an NdFeB magnet (high coercivity, fixed) between steel pole pieces, wound with a copper coil.

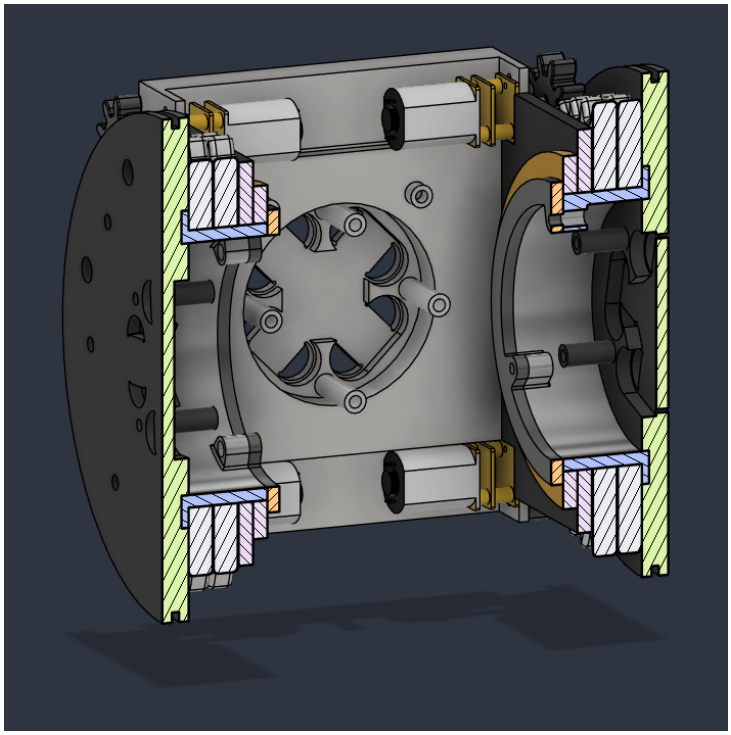


Figure 16: Interior view of the rectangular magnet housing.

Operating principle: A current pulse through the coil reverses AlNiCo 5 magnetisation while NdFeB stays fixed. Fields aligned → **ON** (attract); fields opposed → **OFF** (release). No power needed to hold either state.

Materials: AlNiCo 5 rod, NdFeB magnet, mild steel 1018/S20C pole pieces, 30 AWG enamelled copper wire (200 turns). Original dimensions (4.76 mm dia. × 9.53 mm, four magnets in 42 mm ring) being adapted for the 100 mm face. Final dimensions and driving circuit to be set in Phase B testing.

Additional References

[1] Tosun, T., Davey, J., Liu, C., & Yim, M. (2016). Design and characterization of the EP-Face connector. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (pp. 45–51).